TEACHER GUIDE

OCR GCSE (9-1) in Business (J204)

Financial ratios

5. Finance
5.3 Revenue, costs, profit and loss: profitability ratios



A revision lesson. Pupils run a car dealership in firms of four and revise financial ratios on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Gross profit margin

Round 1

Last year's accounts.

Gross profit margin, as a %?

Cars sold	**12 cars**
Price per car	**£16,500**
Gross profit	**£45,540**

ANSWER

23%

12 × £16,500 = £198,000; £45,540 ÷ £198,000 × 100 = 23%

Why: It keeps 23p of every £1 of sales before overheads.

2 Why net margin fell

Round 1

Gross profit margin stayed the same, but net profit margin fell.

Which two could explain this?

- **A** Cars cost more to buy
- **B** Staff wages rose
- **C** More cars were sold
- **D** Prices were cut
- **E** Rent rose

ANSWER

- ✓ **Staff wages rose**
- ✓ **Rent rose**

Why: Only overheads come between gross and net profit.

3 Rank gross margin

Round 2

Three dealers: sales and gross profit for the year.

Order by gross margin, highest first?

- **A** Sales £500,000, gross profit £100,000
- **B** Sales £200,000, gross profit £50,000
- **C** Sales £80,000, gross profit £24,000

ANSWER

1. **Sales £80,000, gross profit £24,000**
2. **Sales £200,000, gross profit £50,000**
3. **Sales £500,000, gross profit £100,000**

Why: Margins 30%, 25%, 20%: the biggest profit has the lowest margin.

Topic tasks

4 Gross margin change

Round 2

Two years compared.

Gross margin change, in % points?

Sales, last year	**£200,000**
Gross profit, last year	**£32,000**
Sales, this year	**£190,000**
Gross profit, this year	**£47,500**

ANSWER

9%

£47,500 ÷ £190,000 = 25%;
£32,000 ÷ £200,000 = 16%;
25 − 16 = 9 points

Why: The margin went from 16% to 25%.

5 Raise gross margin

Round 3

You want a higher gross profit margin.

Which change works, and why?

A Cut the showroom rent

B Buy cars more cheaply

ANSWER

✔ Buy cars more cheaply

Why: Rent is an overhead; gross margin is price less the cost of the cars.

6 New net margin

Round 3

Overheads are going up.

New net profit margin, as a %?

Sales	**£200,000**
Gross profit	**£19,900**
Overheads	**£9,000**
Overheads rise by	**10%**

ANSWER

5%

£9,000 + 10% = £9,900;
£19,900 − £9,900 = £10,000;
÷ £200,000 = 5%

Why: Work out the new overheads first, then the margin.

Final questions

Each pupil answers 6 final questions on financial ratios. They come from these 12 questions.

1. A dealership has gross profit of £60,000 and sales revenue of £150,000. What is its gross profit margin?
   ✔ **40%**

2. A motor business records net profit of £18,000 and sales revenue of £120,000. What is its net profit margin?
   ✔ **15%**

3. A workshop has £30,000 gross profit and a 30% gross profit margin. What is its sales revenue?
   ✔ **£100,000**

4. A dealership expects £80,000 sales revenue and sets a target net profit margin of 6.25%. What net profit would meet the target?
   ✔ **£5,000**

5. A promoted line's gross profit stays £30,000 while sales revenue rises from £84,000 to £90,000. What is the new gross profit margin?
   ✔ **33.3% (approximately)**

6. A dealer's net profit stays £12,000 while sales revenue rises from £96,000 to £120,000. What happens to net profit margin?
   ✔ **It falls by 2.5 percentage points**

7. The Finance lead divides £140,000 sales revenue by £21,000 net profit and reports a 6.7% net profit margin. What is the correct margin?
   ✔ **15%**

8. Sales revenue rises but net profit margin falls. What should the Finance lead investigate?
   ✔ **Whether costs rose faster than revenue**

9. Gross profit margin = gross profit ÷ revenue × 100. Revenue is £600,000 and gross profit £150,000. Using the formula, what is the gross profit margin?
   ✔ **25%**

10. A dealer's sales revenue is £400,000 and its gross profit £100,000. What is the gross profit margin?
    ✔ **25%**

Final questions

11 A workshop’s gross profit margin is 40% and its sales revenue £250,000. What is its gross profit?

✔ **£100,000**

12 A dealer’s gross profit margin rises from 18% to 22%. What does this show?

✔ **More of each £1 of sales is left after cost of sales**